

# JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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## ABSTRACT

Counterfactual explanations (CE) are most useful when they identify the few input changes needed to obtain a desired model outcome, but many CE methods return counterfactuals that modify more features than necessary. This paper makes three contributions toward sparse, reviewer-verifiable counterfactual refinement. First, we introduce *COLA*, a model- and generator-agnostic post-hoc framework that takes any generated CE and refines it into an action plan with fewer edits. Second, we propose *p-SHAP*, a Shapley attribution method driven by a joint distribution between factual and counterfactual sets; in our main instantiation, this distribution is an optimal-transport (OT) coupling that aligns the two sets before selecting edits. Third, we provide theoretical and empirical evidence for this refinement: OT minimizes an upper bound on the  $W_1$  divergence between factual and counterfactual outcomes, refined counterfactuals are guaranteed under mild conditions not to move farther from factials than the originals, and across four datasets, twelve models, and five CE generators, COLA reaches the same target effects with only 26–45% of the original feature edits. A small-scale benchmark further shows near-optimality.

The experiment code and software will be released upon acceptance.

## 1 BACKGROUND

Explainable Artificial Intelligence (XAI) supports transparent and trustworthy artificial intelligence by explaining how model outputs are produced (Arrieta et al., 2020; Das & Rad, 2020). Two explanation families motivate this paper. Feature Attributions (FA) methods, including Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017), assign importance to input features and therefore help users scan which variables drive a prediction; in healthcare, for example, they can highlight factors such as age and medical history (Ter-Minassian et al., 2023; Nohara et al., 2022). Counterfactual Explanations (CE) methods instead describe how inputs could be changed to obtain another outcome (Wachter et al., 2017; Guidotti, 2022). The CE literature is intentionally diverse rather than universal, with hundreds of algorithms proposed for different assumptions and goals (Guidotti, 2022; Verma et al., 2020): some methods generate one counterfactual per factual instance, some treat multiple factials as a group, some construct global directions for an entire dataset (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024), and others view factials as an empirical distribution and seek a nearby counterfactual distribution (You et al., 2025). Because CE objectives and model assumptions vary across these settings, a contribution of this work is not a new CE generator, but a post-hoc refinement layer that can operate across existing CE generators and model classes.

**Problem Description and Challenges** The central problem is to turn an existing CE into a smaller, clearer action plan without losing the desired counterfactual effect. This question builds on prior work summarized in Appendix A (see also Appendix A for the closest action-cost formulations), but differs from standard CE generation because the input already includes factual data and a counterfactual proposal. It also differs from standalone FA: feature importance alone may identify influential variables without specifying concrete edits, whereas CE alone may prescribe many edits that are unnecessary for the target outcome. We therefore ask the following reviewer-facing question: given factual instance(s) and an available counterfactual proposal, which feature edits are actually needed to preserve the counterfactual effect?

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

| $\mathbf{x}$      | $\mathbf{z}'$     | $\mathbf{z}''$    | $\mathbf{y}^*$    |
|-------------------|-------------------|-------------------|-------------------|
| $\textcircled{Q}$ | $\textcircled{Q}$ | $\textcircled{Q}$ | $\textcircled{R}$ |
| 200               | 250               | 200               | Yes               |
| 150               | 150               | 150               | Yes               |
| 100               | 350               | 100               | Yes               |
| 150               | 150               | 350               | Yes               |
| 5                 | 8                 | 5                 |                   |
| 3                 | 3                 | 7                 |                   |
| 2                 | 9                 | 2                 |                   |
| 6                 | 6                 | 6                 |                   |
| No                | No                | No                |                   |
| No                | Yes               | Yes               |                   |
| No                | No                | No                |                   |
| No                | Yes               | Yes               |                   |

Figure 1: [Example: User engagement on an e-commerce platform] A platform aims to increase user registrations. The platform has collected data on user interactions, such as the amount of money spent ( $\textcircled{Q}$ ), the number of clicks ( $\textcircled{R}$ ), and whether the user has registered ( $\textcircled{R}$ ). In the original data ( $\mathbf{x}$ ), no users are registered. Action plans  $\mathbf{z}'$  and  $\mathbf{z}''$  adjust user characteristics to achieve the desired outcome ( $\mathbf{y}^*$ ) of full registration. Both plans achieve a half counterfactual effect, but  $\mathbf{z}''$  requires fewer modifications compared to  $\mathbf{z}'$ .

Three challenges define the contribution structure. First, the refinement must be CE-generator agnostic, since no single CE algorithm covers instance-based, group-based, and distributional use cases. Second, it must remain model agnostic rather than requiring differentiability or a special model architecture. Third, the attribution used for edit selection must be counterfactual-aware: choosing features from FA scores computed independently of CE can select variables that are globally important but poorly aligned with the path from factual inputs to target outcomes. Result II in the numerical section directly tests this issue by comparing attribution methods that do and do not use factual-counterfactual alignment.

**Main Contributions** The paper contributes a refinement problem, a method, theory, and evidence for sparse counterfactual action plans.

- *Problem framing.* We formulate sparse post-hoc CE refinement as the task in Figure 1: preserve a desired counterfactual effect while limiting the number of changed factual features.
- *Method.* We introduce COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a model- and CE-generator-agnostic framework that uses a joint factual-counterfactual distribution to select feature edits, and we instantiate this idea with  $p$ -SHAP driven by an optimal transport (OT) coupling.
- *Theory.* We show that OT minimizes an upper bound on the  $W_1$  divergence and that, under the stated conditions, the refined counterfactuals are no farther from factuals than the aligned counterfactual proposal.
- *Empirical validation.* Across four datasets, twelve models, and five CE methods, COLA with  $p$ -SHAP achieves the same target effects with only 26–45% of the original feature edits, and a tractable benchmark shows near-optimal behavior.

To our best knowledge, this is the first systematic method for the problem in Figure 1 that does not require fixing a particular CE algorithm or model class.

## 2 PROBLEM FORMULATION

We now make the refinement target precise. Let  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  be a black-box machine learning (ML) model, and let  $\mathbf{x}$  denote observed factual data with  $n$  rows and  $d$  columns ( $\mathbf{x} \in \mathbb{R}^{n \times d}$ ,  $n \geq 1$ , and  $d \geq 1$ ). Let  $\mathbf{y}^*$  be the target model output ( $\mathbf{y}^* \in \mathbb{R}^m$ ,  $m \geq 1$ ). The contribution of this formulation is to expose the action budget explicitly: among all refined counterfactual data matrices  $\mathbf{z}$  ( $\mathbf{z} \in \mathbb{R}^{n \times d}$ ), we seek one whose model output is close to  $\mathbf{y}^*$  while changing at most  $C$  entries of  $\mathbf{x}$ . Let  $D$  be a

divergence measuring dissimilarity between the relevant quantities. The optimization problem is

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The first two lines, equation (1a) and equation (1b), retain the standard CE requirement:  $\mathbf{z}$  should make  $f(\mathbf{z})$  close to  $\mathbf{y}^*$  while remaining close to  $\mathbf{x}$ , so  $\mathbf{z}$  can explain why  $f(\mathbf{x})$  does not achieve  $\mathbf{y}^*$ . The formulation is deliberately divergence-agnostic;  $D$  may be Euclidean distance, OT, maximum mean discrepancy (MMD), or a mean- or median-based discrepancy in model outcomes. The remaining constraints make the paper’s sparse-action contribution visible. On top of the counterfactual data  $\mathbf{z}$ , the binary matrix  $\mathbf{c}$  records which entries may change, equation (1c) limits the total number of changes to  $C$ , and equation (1d)–equation (1e) enforce  $z_{ik} = x_{ik}$  whenever  $c_{ik} = 0$ . When  $c_{ik} = 1$ , the sufficiently large constant  $M$  leaves  $z_{ik}$  free to change.

This formulation explains why attribution is needed: solving equation (1) requires deciding which entries of  $\mathbf{x}$  should receive the scarce edit budget encoded by  $\mathbf{c}$ . The next section reviews Shapley-based FA methods that provide this ranking, and Section 6 later shows why counterfactual-aware attribution is essential for selecting useful edits. The exact problem is computationally difficult even when  $d = 1$  for linear models; Appendix B gives the hardness result.

### 3 PRELIMINARIES ON SHAPLEY VALUE

This section prepares the methodological contribution by reviewing Shapley value methods that can rank candidate feature edits. These baselines are important because COLA is modular: it can use existing Shapley variants, while our proposed  $p$ -SHAP changes how factual and counterfactual rows are aligned before attribution. We first recall the game-theoretic definition and then identify the alignment assumptions made by commonly used feature-attribution variants. A broader overview appears in (Sundararajan & Najmi, 2020).

In cooperative game theory, a coalitional game consists of a finite set of players—features in our setting— $\mathcal{F} = \{1, 2, \dots, d\}$  and a characteristic function  $v$ . The map  $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$  assigns each subset  $S \subseteq \mathcal{F}$  a payoff  $v(S)$ , with  $v(\emptyset) = 0$ . For this paper, the key role of Shapley values is not only to decompose a prediction, but to produce an ordered signal for allocating a limited number of counterfactual edits.

Let  $\Delta(k, S)$  denote the marginal contribution of feature  $k$  to coalition  $S$ . It measures the additional payoff obtained when  $k$  is added to the coalition:

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value  $\phi_i$  for a player  $i$  averages such marginal contributions over all coalitions, as defined in equation (3).

$$\phi_k = \frac{1}{d} \sum_{S \subset \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

The quantity  $\phi_k$  is the normalized average contribution of feature  $k$  across all subsets excluding  $k$ . The binomial term weights coalitions of different sizes, and the final division by  $d$  yields the usual fair allocation of the total payoff across features. In our refinement setting, these values become a scoring mechanism for deciding which row-feature positions should be edited first.

For any data point  $\mathbf{x}_i$ ,  $\mathbf{x}_{i,S}$  denotes the coordinates in  $S$ , and  $\mathbf{x}_{i,\mathcal{F} \setminus S}$  denotes the remaining coordinates; we use the same convention for other vectors. The following variants differ mainly in how they choose the reference values that replace absent features, which is precisely the design choice that  $p$ -SHAP makes explicit.

*Baseline Shapley (B-SHAP)* uses a single baseline point  $\mathbf{r}_j$  as the contrastive reference for  $\mathbf{x}_i$ . Its set function is

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j, \mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

where a feature’s absence is modeled using its value in the reference baseline data point  $\mathbf{r}$  (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). Its implicit requirement is an exact alignment between  $\mathbf{x}$  and  $\mathbf{r}$ .

*Random Baseline Shapley (RB-SHAP)* replaces the fixed reference with an expectation over a background distribution, as used implicitly in (Lundberg & Lee, 2017) and in the widely used “SHAP” library:

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

Here  $\mathcal{D}$  is the background distribution, commonly chosen as the training dataset (Lundberg & Lee, 2017; Merrick & Taly, 2020). This variant removes exact alignment but also does not use CE-specific information.

*Counterfactual Shapley (CF-SHAP)* moves closer to our goal by choosing  $\mathcal{D}$  as a counterfactual distribution conditioned on each factual instance  $\mathbf{x}_i$  ( $i = 1, 2, \dots, n$ ):

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

This method has been adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021) and can improve contrastive FA. Its limitation for our broader contribution is that it assumes a probabilistic alignment between  $\mathbf{x}$  and  $\mathbf{r}$  is already available.

#### 4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY ( $p$ -SHAP) AND ITS THEORETICAL ASPECTS

**(Proposed)  $p$ -SHAP** The first methodological contribution is to make factual-counterfactual alignment an explicit input to Shapley attribution. We generalize equation (4) to equation (6) by integrating an algorithm  $A_{\text{Prob}}$  that returns a joint probability over factual and counterfactual rows.<sup>1</sup> This view turns B-SHAP, RB-SHAP, and CF-SHAP into special cases under different alignment choices. Our  $p$ -SHAP is defined as follows,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

Crucially, unlike prior methods that rely on random baselines,  $p$ -SHAP can solve the alignment problem by identifying an optimal coupling via OT. This reformulates feature attribution not merely as a prediction decomposition, but as a transport-informed mechanism for selecting low-cost explanatory edits.

Remark that  $A_{\text{Prob}}$  is independent of the CE algorithm, but only depends on the generated counterfactual  $\mathbf{r}$ . This independence is central to the paper’s scope:  $p$ -SHAP uses the generated counterfactual set, but it does not require access to the internal distribution or optimization path of the CE generator. That distinction makes the method applicable beyond CF-SHAP settings where a conditional counterfactual distribution is assumed. Result II in Section 6 evaluates the practical consequence: OT can be more effective than relying on a counterfactual distribution defined by a CE generation mechanism as done by Albini et al. (2022).

Our main instantiation chooses  $A_{\text{Prob}}$  by solving the following OT problem, equivalently an entropically regularized 2-Wasserstein objective. The resulting transportation plan  $\mathbf{p}_{\text{OT}}$  supplies the joint distribution between  $\mathbf{x}$  and  $\mathbf{r}$  used by  $p$ -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\mu, \nu)} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left( \frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

<sup>1</sup>First,  $p$ -SHAP degrades to B-SHAP in equation (4) when  $A_{\text{Prob}}$  defines a joint distribution between  $\mathbf{x}$  and  $\mathbf{r}$  that indicates an  $i \leftrightarrow j$  alignment of for any  $\mathbf{x}_i, \mathbf{r}_j$ . Second,  $p$ -SHAP degrades to RB-SHAP in equation (5) when  $A_{\text{Prob}}$  is defined to be independent of CE but associates with an arbitrary distribution  $\mathcal{D}$ . Third,  $p$ -SHAP degrades to CF-SHAP in equation (6) when  $A_{\text{Prob}}$  is built upon a known distribution of CE.

Here  $\mu$  and  $\nu$  are the marginal distributions of  $\mathbf{x}$  and  $\mathbf{r}$ , respectively, and  $\Pi(\mu, \nu)$  is the set of feasible joint distributions, or transport plans. The term  $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$  is entropic regularization with coefficient  $\varepsilon \geq 0$ ; when  $\varepsilon > 0$ , it accelerates OT computation. This coupling is the bridge from counterfactual proposals to actionable feature rankings.

**Theoretical Aspects of  $p$ -SHAP** The theoretical contribution is to connect OT alignment to counterfactual effect. Because OT minimizes the total feature modification cost—that is, modifying  $\mathbf{x}$  toward  $\mathbf{r}$  under an alignment between factual  $\mathbf{x}$  and counterfactual  $\mathbf{r}$ —it directly matches the goal of identifying low-cost modifications from  $\mathbf{x}$  to  $\mathbf{r}$ . Under a Lipschitz continuity assumption on the predictive model  $f$ , Theorem 4.1 below (proof in Appendix C) formalizes this connection: the transportation plan  $\mathbf{p}_{\text{OT}}$  used in  $p$ -SHAP minimizes an upper bound on the divergence between  $f(\mathbf{x})$  and  $\mathbf{y}^*$ . Specifically, the 1-Wasserstein distance between  $f(\mathbf{x})$  and  $\mathbf{y}^*$  is bounded by the Lipschitz constant multiplied by the square root of the minimized expected cost of changing  $\mathbf{x}$  toward  $\mathbf{r}$ , i.e.  $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ , where  $p_{ij}$  ( $j = 1, 2, \dots, m$ ) quantifies how feature values of  $\mathbf{x}_i$  are adjusted toward one or more rows  $\mathbf{r}_j$ . Practically, the OT plan  $\mathbf{p}_{\text{OT}}$  gives  $p$ -SHAP a principled strategy for moving feature values of  $\mathbf{x}$  toward those of  $\mathbf{r}$  while minimizing the expected modification cost  $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ . Compared with other feasible plans ( $\mathbf{p} \in \Pi$ ),  $\mathbf{p}_{\text{OT}}$  provides the smallest such cost and therefore the tightest upper bound, within this analysis, on the counterfactual-effect violation  $W_1(f(\mathbf{x}), \mathbf{y}^*)$ .

**Theorem 4.1** ( $p$ -SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence  $W_1$ , i.e.  $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$ . Suppose the counterfactual outcome  $\mathbf{y}^*$  is fully achieved by  $\mathbf{r}$ , i.e.  $\mathbf{y}_j^* = f(\mathbf{r}_j)$  ( $j = 1, 2, \dots, m$ ). Assume that the model  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  is Lipschitz continuous with Lipschitz constant  $L$ . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by  $\mathbf{p}_{\text{OT}}$  (with  $\varepsilon = 0$ ), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely,  $\mathbf{p}_{\text{OT}}$  minimizes the upper bound of  $W_1(f(\mathbf{x}), \mathbf{y}^*)$ , where the upper bound is based on the expected feature modification cost.

Although Theorem 4.1 bounds transport cost rather than directly solving the  $\mathcal{NP}$ -hard discrete sparsity ( $L_0$ ) problem, it supplies the convex proxy used by COLA. By concentrating attribution mass on efficient factual-counterfactual paths, the OT coupling guides the greedy selection in Algorithm 1 toward edits that discard non-essential features. The second theoretical statement clarifies the attribution interpretation. Theorem 4.2 (proof in Appendix D) states that  $p$ -SHAP attributes the effect of changing characteristics through the corresponding intervention.

**Theorem 4.2** (Interventional Effect of  $p$ -SHAP). *For any subset  $S \subseteq \mathcal{F}$  and any  $\mathbf{x}_i$  ( $i = 1, 2, \dots, n$ ),  $v^{(i)}(S)$  represents the causal effect of the difference between the expected value of  $f(\mathbf{r})$  under the intervention on features  $S$  and the unconditional expected value of  $f(\mathbf{r})$ . Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Furthermore, we remark that  $p$ -SHAP preserves nice axioms of B-SHAP and RB-SHAP, which makes it an effective tool for attributing features. We omit the proof but refer to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) as a reference for axioms of Shapley.

## 5 THE ALGORITHMIC FRAMEWORK COLA

**Sketch** The second methodological contribution is COLA, the wrapper that turns any generated counterfactual set into a sparse action plan. Algorithm 1 targets equation (1) through four modular components: a CE generator, a factual-counterfactual coupling, a Shapley attribution rule, and a value-assignment rule for the selected edits. The modular design is important for review: only the coupling and attribution choice define our proposed instantiation, while the framework can still evaluate alternative baselines. The distinction between  $\mathbf{r}$  and  $\mathbf{z}$  is central:  $\mathbf{r}$  is the generated CE, while

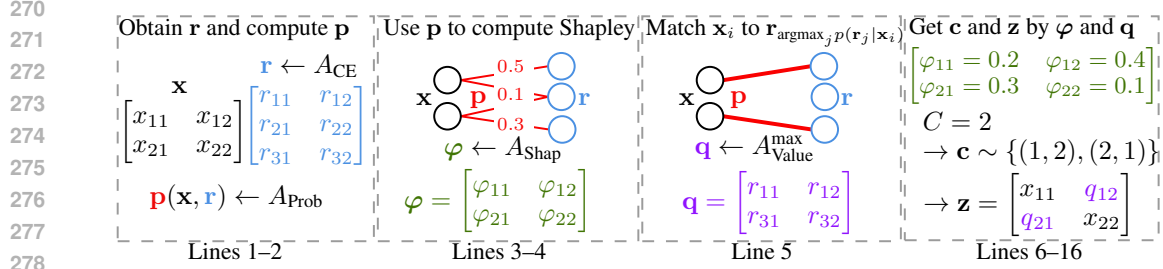


Figure 2: [An illustration of COLA] COLA converts a generated counterfactual into the sparse decision variables of equation (1): the action mask  $\mathbf{c}$  and refined counterfactual  $\mathbf{z}$ . For readability, the illustration uses  $A_{Value}^{max}$  in line 5. In lines 6–16, the budget is  $C = 2$ , so sampling from the attribution-derived probability matrix  $\phi$  selects exactly two feature positions for modification.

$\mathbf{z}$  is the refined CE, and the action mask records how  $\mathbf{z}$  differs from the factual data. We explain Algorithm 1 below, with an illustration in Figure 2; the figure highlights line 5, where the aligned proposal values are inserted into the sparse action plan.

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**Algorithm 1** COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

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**Require:** Model  $f$ , factual  $\mathbf{x} \in \mathbb{R}^{n \times d}$ , target  $\mathbf{y}^* \in \mathbb{R}^m$ ,  $\epsilon$ , and  $C$   
**Ensure:** Action plan  $\mathbf{c} \in \mathbb{R}^{n \times d}$  and correspondingly a refined counterfactual  $\mathbf{z} \in \mathbb{R}^{n \times d}$   
 1: Use  $A_{CE}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$  to obtain  $\mathbf{r} \in \mathbb{R}^{m \times d}$ , with  $\mathbf{y}^* = f(\mathbf{r})$  and  $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$ .  
 2: Use  $A_{Prob}(\mathbf{x}, \mathbf{r})$  to obtain the joint distribution matrix  $\mathbf{p} \in \mathbb{R}_+^{n \times m}$ .  
 3: Use  $A_{Shap}(\mathbf{x}, \mathbf{r}, \mathbf{p})$  to obtain the shapley values  $\phi \in \mathbb{R}^{n \times d}$  for each element of  $\mathbf{x}$ .  
 4: Normalize the element-wise absolute values of  $\phi$ , i.e.,  $\phi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$  ( $\phi \in \mathbb{R}_+^{n \times d}$ ).  
 5: Use  $A_{Value}(\mathbf{r}, \mathbf{p})$  to obtain matrix  $\mathbf{q} \in \mathbb{R}^{n \times d}$ .  
 6: For  $\mathbf{c} \in \{0, 1\}^{n \times d}$ ,  $c_{ik} \leftarrow 0$  ( $i = 1, \dots, n$ ,  $k = 1, \dots, d$ ).  
 7: Sample  $C$  pairs  $(i, k)$  according to the probability matrix  $\phi$ , and let  $c_{ik} = 1$  for them.  
 8: Let  $\mathbf{z} \leftarrow \mathbf{x}$  ( $\mathbf{z} \in \mathbb{R}^{n \times d}$ ).  
 9: **for**  $i \leftarrow 1$  to  $n$  **do**  
 10:   **for**  $k \leftarrow 1$  to  $d$  **do**  
 11:     **if**  $c_{ik} = 1$  **then**  
 12:        $z_{ik} \leftarrow q_{ik}$   
 13:     **end if**  
 14:   **end for**  
 15: **end for**  
 16: **return**  $\mathbf{c}$  and  $\mathbf{z}$

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**Line 1** (Applying a CE algorithm to find a counterfactual  $\mathbf{r}$ ). COLA begins with an existing CE method rather than replacing it. The algorithm  $A_{CE}$  takes  $f$ ,  $\mathbf{x}$ ,  $\mathbf{y}^*$ , and  $\epsilon$  as input and returns a counterfactual  $\mathbf{r}$  close to  $\mathbf{x}$ , with  $\mathbf{y}^* = f(\mathbf{r})$  and  $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$ . This line is what makes the framework CE-generator agnostic while keeping the original factual and counterfactual objects explicit.

**Line 2** (Seeking a joint distribution of  $\mathbf{x}$  and  $\mathbf{r}$ ). The coupling step is the alignment interface of the framework.  $A_{Prob}$  takes  $\mathbf{x}$  and  $\mathbf{r}$  and outputs a matrix  $\mathbf{p}$  that represents a joint distribution over the  $n$  rows in  $\mathbf{x}$  and the  $m$  rows in  $\mathbf{r}$ . COLA later uses this same alignment both to compute candidate replacement values for  $\mathbf{z}$  and, in  $p$ -SHAP, to compute counterfactual-aware attributions. When the true row alignment between  $\mathbf{x}$  and  $\mathbf{r}$  is unknown, Section 3 motivates using OT; when reliable alignment information is available,  $A_{Prob}$  can encode it directly.

**Lines 3–4** ( $p$ -SHAP FA). We apply equation (7) to compute Shapley values for  $\mathbf{x}$ . The joint distribution  $\mathbf{p}$  is the key contribution carried into this line: it aligns each row of  $\mathbf{x}$  with representative counterfactual rows of  $\mathbf{r}$ , so the attribution scores rank edits relative to the target counterfactual effect rather than relative to an unrelated baseline. The element-wise absolute Shapley values of  $\mathbf{x}$  are then normalized to sum to one, producing  $\phi$  as the probability matrix used for sparse edit selection.

**Line 5** (*Computing feature values*). The value-assignment step separates the question “which entries should change?” from “what values should they take?” Given  $\mathbf{r}$  and  $\mathbf{p}$ ,  $A_{\text{Value}}$  selects one or more counterfactual rows as references for each factual row and returns  $\mathbf{q} \in \mathbb{R}^{n \times d}$ , where  $q_{ik}$  is the candidate replacement for  $x_{ik}$ . We use two simple instantiations:  $A_{\text{Value}}^{\max}$  for a single most-aligned row and  $A_{\text{Value}}^{\text{avg}}$  for a weighted average over multiple rows.

For any row  $\mathbf{x}_i$ ,  $A_{\text{Value}}^{\max}$  selects the row of  $\mathbf{r}$  with the highest coupling probability.

$$\mathbf{q} = A_{\text{Value}}^{\max}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i),k} \text{ and } \tau(i) = \arg \max_{j=1,2,\dots,m} p_{ij}. \quad (9)$$

The algorithm  $A_{\text{Value}}^{\text{avg}}$  instead computes  $q_{ik}$  as a convex combination, i.e., a weighted average of  $r_{1k}, r_{2k}, \dots, r_{mk}$ .

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left( \frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

**Lines 6–16.** Because  $\varphi$  is non-negative and normalized, COLA treats it as a policy over row-feature positions in  $\mathbf{x}$ . Line 7 samples  $C$  positions, sets  $c_{ik} = 1$  for those positions, and replaces the corresponding  $x_{ik}$  values by  $q_{ik}$ . The returned pair  $(\mathbf{c}, \mathbf{z})$  is therefore directly interpretable as the sparse action plan and its refined counterfactual realization for equation (1).

By Theorem 4.1, COLA is designed to reduce the dissimilarity between  $f(\mathbf{z})$  and  $\mathbf{y}^*$  by spending the edit budget on attribution-selected positions that are most relevant to the desired counterfactual effect. The next guarantee adds the proximity side of the contribution. The counterfactual-proximity theorem (proof in Appendix E) shows that the refined  $\mathbf{z}$  produced by COLA does not move farther from  $\mathbf{x}$  than the aligned counterfactual proposal under the Frobenius norm. Table 3 provides empirical evidence consistent with this proximity behavior across the experimental settings.

**Theorem 5.1** (Counterfactual Proximity). *Let  $\mathbf{q} \in \mathbb{R}^{n \times d}$  be the aligned reference matrix derived from the counterfactual set  $\mathbf{r} \in \mathbb{R}^{m \times d}$  via the value assignment step  $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ , where  $\mathbf{p}$  is the coupling matrix. For any dimensions  $n, m \geq 1$ , the refined counterfactual  $\mathbf{z}$  constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

*This inequality guarantees that the refined counterfactual  $\mathbf{z}$  is strictly closer (or equal) to the factual input  $\mathbf{x}$  compared to the aligned counterfactual proposal  $\mathbf{q}$ , ensuring that the refinement process introduces no additional divergence.*

**Corollary 5.2.** *In the special case where  $n = m$  and the OT plan  $\mathbf{p}$  corresponds to a deterministic permutation  $\sigma$  (i.e., obtained without entropic regularization,  $\varepsilon = 0$ ), the matrix  $\mathbf{q}$  becomes a row-permuted version of  $\mathbf{r}$  (denoted as  $\mathbf{r}_\sigma$ , where  $q_i = r_{\sigma(i)}$ ). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

*recovering the intuition that the refined counterfactual is at least as close to the facts as the original counterfactual set reordered by  $\sigma$ .*

**Complexity of COLA** The computational contribution of COLA is that it adds a modular refinement cost on top of the chosen CE generator. Let  $\mathcal{O}(M_{\text{CE}})$  be the complexity of  $A_{\text{CE}}$ . For  $A_{\text{Shap}}$ , consider weighted linear regression to estimate Shapley values, and let  $M_{\text{Shap}}$  be the number of sampled subsets. With  $n$  factual rows,  $m$  counterfactual rows,  $d$  features, and entropic-OT regularization parameter  $\varepsilon$ , the total complexity is  $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$  where  $N = \mathcal{O}(nm) + \mathcal{O}(nd)$  for  $A_{\text{Value}}^{\max}$  and  $N = \mathcal{O}(nmd)$  for  $A_{\text{Value}}^{\text{avg}}$ . Appendix F gives the detailed derivation.

## 6 NUMERICAL RESULTS

This section tests the three empirical claims that correspond to the paper’s contributions. We evaluate COLA on equation (1) with  $\mathbf{y}^* = f(\mathbf{r})$ , where  $\mathbf{r}$  is produced by a CE method  $A_{\text{CE}}$ , and we track the feasibility role of equation (1b), the action budget in equation (1c), and the edit constraints through equation (1e). To avoid tying the contribution to one discrepancy measure, we report four divergences: OT for whole-distribution distance, MMD for high-dimensional mean discrepancy, absolute mean difference (MeanD) for mean difference, and absolute median difference (MedianD) for median difference. The experiments are organized to show: I) COLA’s effectiveness for modification minimality. II) p-SHAP’s superior performance over other Shapley methods for sparse counterfactual refinement. III) COLA’s near-optimal performance on tractable benchmarks.

**Experiment Setup** The benchmark is designed to make the generality claim visible. We use 4 binary-classification datasets, 5 CE algorithms with different objectives, and 12 classifiers, summarized in Table 1. A single “experiment scenario” is a dataset, an  $A_{CE}$  algorithm, and a model; throughout the section, references to Table 1 identify this scenario grid, and Table 1 remains the compact map of all benchmark ingredients.

Table 1: [Experiment Scenarios Setup] Evaluation scope for the model- and generator-agnostic contribution: four datasets, five CE algorithms, and 12 models. A “scenario” is one combination of dataset,  $A_{CE}$  algorithm, and model  $f$ .

|                             |                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dataset</b>              | HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)                                                                                                                                                          |
| $A_{CE}$                    | DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)                                                                                                |
| <b>Model <math>f</math></b> | Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA) |

Table 2: [Experiment Methods Setup] Six attribution-and-coupling variants isolate the contribution of counterfactual information and OT alignment. Colors match Figure 3; each method is evaluated within the scenarios defined in Table 1.

| Method               | The probability $\mathbf{p}$ used by $A_{\text{value}}$ and $A_{\text{Shap}}$                                        | The Shapley algorithm $A_{\text{Shap}}$                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| RB- $p_{\text{Uni}}$ | $\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$                                              | RB-SHAP, $\mathcal{D} = \text{trainset of } f$                       |
| RB- $p_{\text{OT}}$  | $\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$                  | RB-SHAP, $\mathcal{D} = \text{trainset of } f$                       |
| CF- $p_{\text{Uni}}$ | $\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$                                              | CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$                  |
| CF- $p_{\text{Rnd}}$ | $\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$       | CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$                  |
| CF- $p_{\text{OT}}$  | $\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$                                                     | $p$ -SHAP with $\mathbf{p}$                                          |
| CF- $p_{\text{Ect}}$ | $\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$ | (CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$ |

The selected  $A_{CE}$  methods span the use cases targeted by COLA. DiCE and KNN are instance-based methods that generate counterfactuals for individual factual instances. AReS and GlobeCE are group-based methods that produce counterfactual instances for the factual data as a group. Discount treats factual instances as an empirical distribution and seeks a nearby counterfactual distribution.

Table 2 defines six methods. CF- $p_{\text{OT}}$  is the proposed  $p$ -SHAP instantiation, while the other rows are baselines that remove or alter parts of the contribution. Each method is specified by  $A_{\text{Prob}}$ ,  $A_{\text{value}}$ , and  $A_{\text{Shap}}$ . For example, RB- $p_{\text{Uni}}$  uses a uniform  $A_{\text{Prob}}$  to compute  $\mathbf{p}$ , passes that  $\mathbf{p}$  into equation (9) or equation (10) to compute  $\mathbf{q}$ , and uses RB-SHAP as  $A_{\text{Shap}}$ .

These variants form the ablation logic. RB- $p_{\text{Uni}}$  versus CF- $p_{\text{Uni}}$  tests whether CE information matters. CF- $p_{\text{Uni}}$ , CF- $p_{\text{Rnd}}$ , and CF- $p_{\text{OT}}$  test whether the particular joint distribution matters once CE information is used. CF- $p_{\text{OT}}$  versus RB- $p_{\text{OT}}$  isolates whether OT helps through Shapley alignment rather than merely through value assignment. Finally, CF- $p_{\text{Ect}}$  represents the special case where each counterfactual has a known factual source, so the exact alignment makes CF-SHAP coincide with B-SHAP.

## Result I: COLA achieves significant action reduction with a minor loss in counterfactual effect

This result supports the core sparse-refinement contribution. In Table 3, the target is for  $\mathbf{z}$  to satisfy  $f(\mathbf{z})$  reaching either 80% or 100% of the counterfactual effect of  $f(\mathbf{r})^2$ . The table shows that COLA reaches these targets with far fewer actions than the original CE  $\mathbf{r}$ . For the 80% effect target, COLA uses only 13%–25% of the original feature-change magnitude, measured by  $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$ . For the stricter 100% target, only COLA with  $p$ -SHAP (CF- $p_{\text{OT}}$ ) succeeds across the reported datasets, using

<sup>2</sup>Note that by definition, we have  $D(f(\mathbf{r}), \mathbf{y}^*) = 0$ , which represents a 100% counterfactual effect since  $\mathbf{y}^* = f(\mathbf{r})$ . To define a counterfactual effect 80%, consider the proportion of divergence reduced by the refined CE  $\mathbf{z}$ . That is,  $\text{Counterfactual Effect} = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$ , with  $D$  being OT.

Table 3: [COLA for Modification Minimality] Sparse-refinement summary: the table reports how many features each method modifies in  $\mathbf{z}$  when  $f(\mathbf{z})$  reaches 80% or 100% of the counterfactual effect of  $f(\mathbf{r})$ . Each entry averages 4 randomly selected scenarios from Table 1, with 100 runs per scenario. The symbol “-” means the target counterfactual effect is not achieved.

| Dataset                                | Method               | 80% Counterfactual Effect          |                                                               | 100% Counterfactual Effect          |                                                               |
|----------------------------------------|----------------------|------------------------------------|---------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------|
|                                        |                      | # Modified Features                | $\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$ | # Modified Features                 | $\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$ |
| German Credit<br>$ \mathcal{F}  = 9$   | RB- $p_{\text{Uni}}$ | —                                  | —                                                             | —                                   | —                                                             |
|                                        | RB- $p_{\text{OT}}$  | 5.29( $\pm 0.09$ )                 | 75.9%                                                         | —                                   | —                                                             |
|                                        | CF- $p_{\text{Uni}}$ | —                                  | —                                                             | —                                   | —                                                             |
|                                        | CF- $p_{\text{Rnd}}$ | —                                  | —                                                             | —                                   | —                                                             |
|                                        | CF- $p_{\text{OT}}$  | <b>1.70(<math>\pm 0.02</math>)</b> | <b>24.3%</b>                                                  | <b>3.13(<math>\pm 0.03</math>)</b>  | <b>44.9%</b>                                                  |
| Hotel Bookings<br>$ \mathcal{F}  = 29$ | RB- $p_{\text{Uni}}$ | 7.10( $\pm 0.08$ )                 | 24.5%                                                         | —                                   | —                                                             |
|                                        | RB- $p_{\text{OT}}$  | 8.55( $\pm 0.08$ )                 | 50.2%                                                         | —                                   | —                                                             |
|                                        | CF- $p_{\text{Uni}}$ | 7.01( $\pm 0.07$ )                 | 41.1%                                                         | —                                   | —                                                             |
|                                        | CF- $p_{\text{Rnd}}$ | 10.63( $\pm 0.08$ )                | 62.4%                                                         | —                                   | —                                                             |
|                                        | CF- $p_{\text{OT}}$  | <b>2.50(<math>\pm 0.03</math>)</b> | <b>14.6%</b>                                                  | <b>4.44(<math>\pm 0.02</math>)</b>  | <b>26.0%</b>                                                  |
| COMPAS<br>$ \mathcal{F}  = 15$         | RB- $p_{\text{Uni}}$ | 5.02( $\pm 0.05$ )                 | 82.7%                                                         | —                                   | —                                                             |
|                                        | RB- $p_{\text{OT}}$  | —                                  | —                                                             | —                                   | —                                                             |
|                                        | CF- $p_{\text{Uni}}$ | 2.80( $\pm 0.04$ )                 | 34.4%                                                         | —                                   | —                                                             |
|                                        | CF- $p_{\text{Rnd}}$ | 2.58( $\pm 0.04$ )                 | 32.1%                                                         | —                                   | —                                                             |
|                                        | CF- $p_{\text{OT}}$  | <b>1.25(<math>\pm 0.03</math>)</b> | <b>14.8%</b>                                                  | <b>2.45(<math>\pm 0.03</math>)</b>  | <b>30.0%</b>                                                  |
| HELOC<br>$ \mathcal{F}  = 23$          | RB- $p_{\text{Uni}}$ | —                                  | —                                                             | —                                   | —                                                             |
|                                        | RB- $p_{\text{OT}}$  | —                                  | —                                                             | —                                   | —                                                             |
|                                        | CF- $p_{\text{Uni}}$ | —                                  | —                                                             | —                                   | —                                                             |
|                                        | CF- $p_{\text{Rnd}}$ | 2.73( $\pm 0.04$ )                 | 15.7%                                                         | —                                   | —                                                             |
|                                        | CF- $p_{\text{OT}}$  | <b>2.35(<math>\pm 0.03</math>)</b> | <b>13.4%</b>                                                  | <b>7.745(<math>\pm 0.05</math>)</b> | <b>44.7%</b>                                                  |

only 26%–45% of the feature changes in the original  $\mathbf{r}$  while producing refined  $\mathbf{z}$ . Thus the main empirical message is not simply that COLA changes fewer features, but that OT-aligned  $p$ -SHAP preserves the counterfactual effect while doing so.

### Result II: $p$ -SHAP outperforms the other Shapley methods in achieving counterfactual effect

This result isolates the  $p$ -SHAP contribution. Figure 3 compares the Shapley methods in equation (4)–equation (7), with action budget  $C$  on the x-axis and  $D(f(\mathbf{z}), \mathbf{y}^*)$  from equation (1) on the y-axis. Three comparisons are most informative. First, RB- $p_{\text{Uni}}$  and RB- $p_{\text{OT}}$  perform substantially worse than the counterfactual-aware methods, showing that sparse CE refinement needs CE information in the attribution step. Second, CF- $p_{\text{OT}}$  outperforms RB- $p_{\text{OT}}$ ; because these variants differ in  $A_{\text{Shap}}$  (see Table 2), the gain is attributable to using OT for factual-counterfactual alignment in Shapley attribution. Third,  $p$ -SHAP outperforms CF- $p_{\text{Uni}}$  and CF- $p_{\text{Rnd}}$ , demonstrating that CE information alone is insufficient: the joint distribution must align factual and counterfactual rows effectively. As Figure 4 later suggests, the best alignment need not be the exact pairing implied by the CE generation mechanism.

Across the scenarios in Table 1, COLA with  $p$ -SHAP consistently provides strong sparse-refinement performance, as shown by both Figure 3 and Table 3. Appendix H reports additional scenarios. The reviewer-facing conclusion is that the proposed OT-informed attribution is the strongest tested Shapley choice for sparse CE refinement.

### Result III: COLA is competitive with the mixed integer linear programming (MILP) optimum on tractable benchmarks

This result asks whether the proposed heuristic refinement is close to an exact optimum when such an optimum can be computed. It also probes whether  $p$ -SHAP can reduce dependence on CE-generation-specific knowledge<sup>3</sup> by replacing that knowledge with the OT joint distribution between factual and counterfactual data, as shown in Figure 4. We benchmark CF- $p_{\text{Ect}}$  within COLA and solve equation (1) with MeanD as  $D$ . Because CF- $p_{\text{Ect}}$  uses known factual-counterfactual alignment, it provides a useful comparison point for alignment quality. The

<sup>3</sup>This knowledge, i.e. exact alignment between factual and counterfactual, is available only by DiCE and KNN among all CE methods considered.

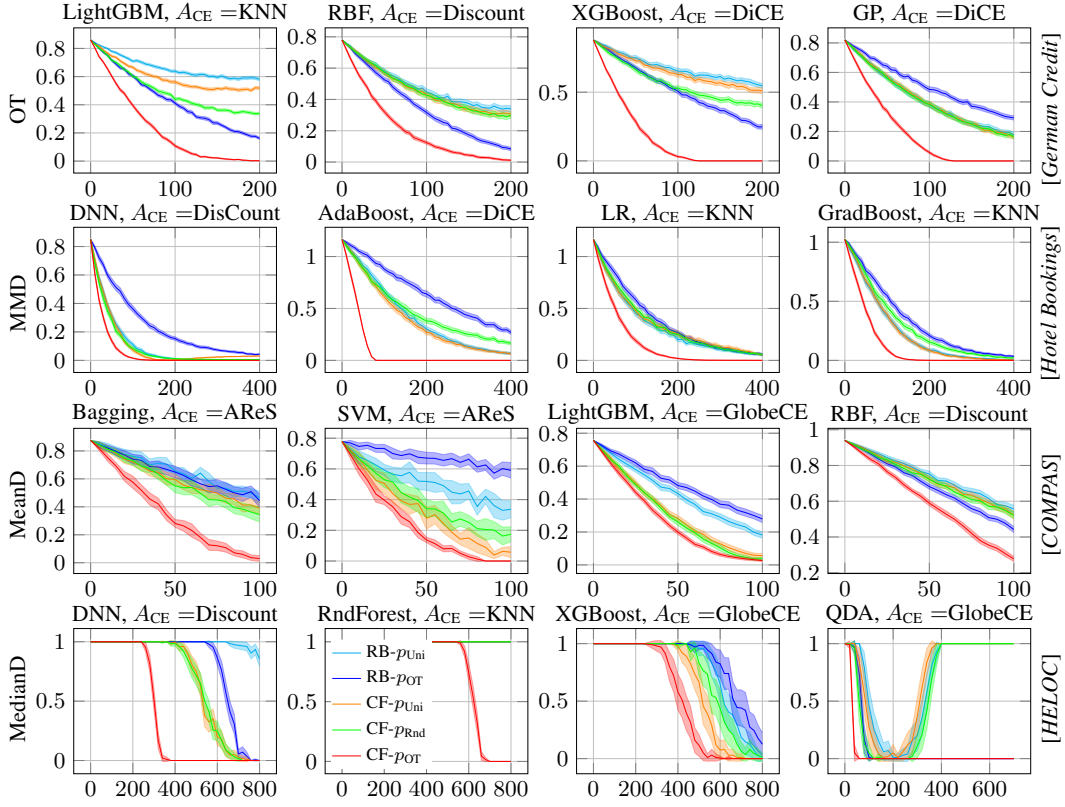


Figure 3: Counterfactual-effect curves for the attribution contribution:  $D(f(\mathbf{z}), \mathbf{y}^*)$  is plotted against the allowed action budget  $C$ . Each experiment uses 100 runs, and shaded regions show 99.9% confidence intervals.  $A_{\text{Value}}^{\text{avg}}$  is used for *HELOC* and *COMPAS*, and  $A_{\text{Value}}^{\text{max}}$  is used for *German Credit* and *Hotel Bookings*. The legend inside “RndForest,  $A_{\text{CE}} = \text{KNN}$ ” applies to all plots.

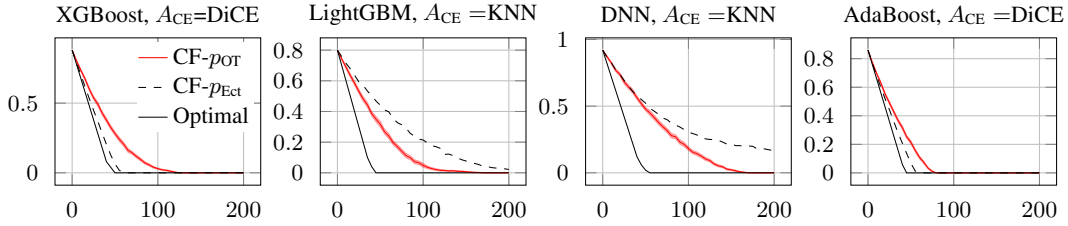


Figure 4: [*German Credit*] Near-optimality benchmark for the tractable case:  $D(f(\mathbf{z}), \mathbf{y}^*)$  versus allowed actions  $C$ , with  $D$  set to *MeanD*.

exact optimum in this setting is obtained via an MILP; Appendix G derives the formulation. Since solving the MILP is computationally expensive, this benchmark is limited to *German Credit*.

CF- $p_{\text{Ect}}$  is near-optimal with DiCE. Because DiCE and KNN generate independent factual-counterfactual pairs, this result supports COLA for instance-based CE even though equation (1) also covers group and distributional CE. Notably, CF- $p_{\text{OT}}$  sometimes outperforms CF- $p_{\text{Ect}}$ , because it uses a more theoretically grounded alignment to identify features requiring modification, whereas CF- $p_{\text{Ect}}$  relies on CE-algorithm-dependent knowledge whose usefulness for FA is not guaranteed. At the same time, KNN still shows a gap between CF- $p_{\text{Ect}}$ , CF- $p_{\text{OT}}$ , and the optimum, so identifying the best alignment remains an open question rather than a settled claim of this paper.

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## 7 CONCLUSIONS

This paper contributes a post-hoc refinement framework for making counterfactual explanations more actionable with fewer edits. The central methodological contribution is COLA, which refines existing CE outputs through joint-distribution-informed Shapley values rather than replacing the CE generator. The central attribution contribution is  $p$ -SHAP, which uses factual-counterfactual alignment—instantiated by OT in our main method—to decide where the limited action budget should be spent. The theory shows that OT minimizes an upper bound relevant to counterfactual effect and that the COLA refinement does not move farther from factuality than its aligned counterfactual proposal. The experiments show the practical salience of these contributions: across four datasets, twelve models, and five CE generators, COLA with  $p$ -SHAP preserves target effects using only 26–45% of the original feature edits, and a tractable benchmark indicates near-optimal behavior. The remaining gap to the optimum in some KNN settings highlights the main limitation and future direction: better factual-counterfactual alignment may further improve sparse counterfactual refinement.

## ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.<sup>4</sup> The contribution of this paper is methodological: *COLA* with *p*-SHAP refines existing CE into smaller action plans. The work is empirical and algorithmic; it does not involve human subjects or interventions, and we did not collect, annotate, or release new personal data.

All experiments use standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC). We use these data only to benchmark the refinement contribution, follow dataset licenses and terms, and do not attempt re-identification or linkage. We release no additional personal information. Our code will include options to exclude protected attributes and enforce feasibility or immutability constraints on features.

CE can reveal biases in underlying models, but they do not by themselves guarantee fairness. Our method reduces the number of recommended feature edits; it does *not* authorize edits to sensitive or immutable attributes (e.g., sex, race, age), and it does not establish causal validity. We discourage using CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals toward unrealistic or unethical behavior changes. Any deployment in consequential domains should add domain-expert feasibility constraints, fairness audits (e.g., disparate impact or benefit across groups), human oversight, and compliance with applicable data-protection and AI regulations. We explicitly caution that our attribution is distributional rather than causal and must not be interpreted as a causal effect.

The main risks are (i) interpreting attributions as causal, (ii) recommending infeasible or harmful edits, and (iii) enabling strategic manipulation. We mitigate these risks by stating the non-causal scope of the contribution, supporting cost and feasibility masks and immutable features in the implementation, reporting subgroup performance where relevant, and documenting responsible-use guidance.

We provide an anonymized implementation, detailed settings, and seeds so the empirical contribution can be replicated. We report model types, hyperparameters, and compute budgets, and we avoid result cherry-picking by using fixed evaluation protocols. To our knowledge, there are no conflicts of interest or sensitive sponsorships related to this submission; any funding disclosures will be provided after the double-blind review.

## REPRODUCIBILITY STATEMENT

We make each contribution traceable to reproducible artifacts. The COLA framework and  $p$ -SHAP attribution are specified with notation and pseudocode in the main paper (Algorithm 1 and the Method and Preliminaries sections), while all theoretical assumptions and complete proofs are provided in the appendix.

Appendix I documents hyperparameters, optimization settings, environment specifications, random seeds, datasets, data splits, preprocessing, hardware, and runtime; Appendix I is therefore the

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<sup>4</sup><https://iclr.cc/public/CodeOfEthics>

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main checklist for reproducing the experiments. The experimental sections enumerate the models, CE generators, coupling choices, and ablations, with configurations mirrored in the supplemental materials. The experiment code and software will be released upon acceptance, and the appendix includes the ablation studies and small-scale MILP benchmark setup needed to independently verify the main empirical claims.

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## APPENDICES

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### A COMPARISON TO EXISTING APPROACHES

Existing work motivates but does not cover the contribution of sparse, post-hoc CE refinement. CF-SHAP (Albini et al., 2022) uses counterfactual data points as the Shapley background distribution, but it assumes that the counterfactual distribution is conditionally defined for each factual instance (Albini et al., 2022). This assumption implies a known probabilistic alignment between factual and counterfactual instances; related assumptions also appear in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). Such alignment is unavailable in many global, group, or distributional explanation settings, so the setup in (Albini et al., 2022) is best viewed as a special case of our joint-distribution formulation. WeightedSHAP (Kwon & Zou, 2022) weights feature contributions, whereas our method weights data-point relationships and can be extended to combine row and column weighting. Work that combines FA and CE for explainability (Kommiya Mothilal et al., 2021) still inherits the same alignment limitation, making it difficult to apply directly to group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025). Another related direction models counterfactuals through causal structure and time-series predictors to decompose a single system-level change (Sharma et al.). In contrast, the contribution here is the minimal-action CE problem: refine already generated counterfactuals so the target effect is preserved with fewer feature edits.

The action-cost problems in (Kanamori et al., 2022; Karimi et al., 2021) are closest to our formulation. (Karimi et al., 2021) minimizes the cost of actions under an assumed known structural causal model (SCM), which is rarely available in practice. (Kanamori et al., 2022) notes that whether existing CE methods can solve this kind of problem remains open, and proposes an MILP-solver approach for linear classifiers, tree ensembles, and deep ReLU networks building on (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). Our contribution differs by treating the existing CE output as input and providing a scalable, model-agnostic refinement layer; the MILP remains useful in this paper only as a small-scale optimality benchmark because exact solving is costly.

### B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 explains why the paper contributes a practical refinement framework rather than a scalable exact solver. Unless  $\mathcal{P} = \mathcal{NP}$ , one should not expect a generally scalable exact algorithm; the difficulty comes not only from possible nonlinearity of  $f$ , but also from the combinatorial edit-selection structure.

**Theorem B.1.** *Problem equation (1) is generally  $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when  $d = 1$  for linear models.*

*Proof.* We reduce from *Sparse Regression (SR) with a Cardinality Constraint*. Given a matrix  $\mathbf{W} \in \mathbb{R}^{m \times n}$ , a target vector  $\mathbf{y}^* \in \mathbb{R}^m$ , and a sparsity level  $K \in \mathbb{N}$ , this problem asks for a vector

$\mathbf{z} \in \mathbb{R}^n$  that minimizes residual error while using at most  $K$  non-zero entries:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

Here  $\|\mathbf{z}\|_0$  counts non-zero entries. The problem is known to be  $\mathcal{NP}$ -hard because selecting the active subset of variables is combinatorial.

We map this SR instance into Problem equation (1). Set the number of features to  $d = 1$  and the number of instances to  $n$ , matching the dimension of the sparse-regression variable. Let the factual data be  $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$  (the zero vector), let the target model output be the same  $\mathbf{y}^* \in \mathbb{R}^m$ , and define the linear model  $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$ .

For model outputs, define  $D(f(\mathbf{z}), \mathbf{y}^*) = \|f(\mathbf{z}) - \mathbf{y}^*\|_2^2$ . For instances, define  $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$ , since  $\mathbf{x} = \mathbf{0}$ . Set the edit budget  $C$  equal to  $k$ , the sparsity level in the sparse-regression problem, and choose any sufficiently large positive  $M$ , for example  $M \geq \max_i |z_i|$ .

Given that  $\mathbf{x} = \mathbf{0}$ , constraints equation (1d) and equation (1e) are simplified to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

Thus  $c_i = 0$  forces  $z_i = 0$ , while  $c_i = 1$  permits  $z_i \in [-M, M]$ . The budget constraint equation (1c) becomes  $\sum_{k=1}^n c_i \leq k$ , exactly matching the sparsity budget.

Our problem equation (1) thus becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this construction,  $c_i$  marks whether  $z_i$  may depart from zero. The constraints therefore reproduce the sparse-regression cardinality constraint inside the CE-refinement formulation, and the objective is identical to the sparse-regression objective.

Because this special case of Problem equation (1) directly encodes cardinality-constrained SR, solving the refinement problem is at least as hard as solving SR. Hence Problem equation (1) is  $\mathcal{NP}$ -hard even for  $d = 1$ , linear  $f$ , and standard Euclidean divergences. This supports the paper's focus on scalable, attribution-guided refinement.  $\square$

## C PROOF OF THEOREM 4.1: $p$ -SHAP TOWARDS COUNTERFACTUAL EFFECT

This proof supports the theoretical contribution that OT alignment is not an arbitrary design choice: it minimizes an upper bound tied to counterfactual effect. Let  $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$  be factual data with probability weights  $\mu_i \geq 0$  satisfying  $\sum_{i=1}^n \mu_i = 1$ , and let  $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$  be counterfactual data with weights  $\nu_j \geq 0$  satisfying  $\sum_{j=1}^m \nu_j = 1$ . Let  $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$  be the OT plan between  $\mathbf{x}$  and  $\mathbf{r}$  that minimizes expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where  $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$  is the set of joint distributions satisfying the marginal constraints  $\sum_{j=1}^m p_{ij} = \mu_i$  for all  $i$  and  $\sum_{i=1}^n p_{ij} = \nu_j$  for all  $j$ . The theorem below shows why this plan is useful for COLA: it aligns attribution with the expected cost of feature modifications, which is the quantity the sparse action plan should conserve.

**Theorem C.1** (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence  $W_1$ , i.e.  $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$ . Suppose the counterfactual outcome  $\mathbf{y}^*$  is fully achieved by  $\mathbf{r}$ , i.e.  $\mathbf{y}_j^* = f(\mathbf{r}_j)$  ( $j = 1, 2, \dots, m$ ). Assume that the model  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  is Lipschitz continuous with Lipschitz constant  $L$ . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by  $\mathbf{p}_{OT}$  (with  $\varepsilon = 0$ ), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Equivalently,  $\mathbf{p}_{OT}$  minimizes the expected-cost upper bound on  $W_1(f(\mathbf{x}), \mathbf{y}^*)$ .

*Proof.* Because  $f$  is Lipschitz continuous with constant  $L$ , every factual-counterfactual pair satisfies:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying by the non-negative OT mass  $p_{ij}^{OT}$  preserves the inequality:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing over all factual and counterfactual rows yields the corresponding expected bound:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let  $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$  and  $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$ . The preceding inequality becomes:

$$E_f \leq L E_d.$$

To express this in terms of squared transportation cost, we apply Cauchy-Schwarz. The weights  $p_{ij}^{OT}$  are non-negative and sum to one because  $\mathbf{p}_{OT}$  is a joint probability distribution over  $\mathbf{x}$  and  $\mathbf{r}$ . For any real-valued functions  $a_{ij}$  and  $b_{ij}$ ,

$$\left( \sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left( \sum_{i,j} a_{ij}^2 \right) \left( \sum_{i,j} b_{ij}^2 \right).$$

Set  $a_{ij} = \sqrt{p_{ij}^{OT}}$  and  $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$ . Then

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy-Schwarz inequality:

$$E_d^2 \leq \left( \sum_{i,j} a_{ij}^2 \right) \left( \sum_{i,j} b_{ij}^2 \right) = \left( \sum_{i,j} p_{ij}^{OT} \right) \left( \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since  $\sum_{i,j} p_{ij}^{OT} = 1$ , this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting this distance bound into the inequality for  $E_f$  gives

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than  $E_f^5$ , then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Thus the 1-Wasserstein divergence between  $f(\mathbf{x})$  and  $\mathbf{y}^*$  is bounded by  $L$  times the square root of the expected transportation cost under  $\mathbf{p}_{\text{OT}}$ . Because  $\mathbf{p}_{\text{OT}}$  minimizes  $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$  over all feasible transport plans in  $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ ,

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

This proves the claimed bound and shows why the OT coupling is the cost-minimizing alignment for the theoretical proxy used by  $p$ -SHAP.  $\square$

## D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

**Theorem D.1** (Theorem 4.2 in the main text). *For any subset  $S \subseteq \mathcal{F}$  and any  $\mathbf{x}_i$  ( $i = 1, 2, \dots, n$ ),  $v^{(i)}(S)$  represents the causal effect of the difference between the expected value of  $f(\mathbf{r})$  under the intervention on features  $S$  and the unconditional expected value of  $f(\mathbf{r})$ . Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

*Proof.* Let  $p(\mathbf{x}, \mathbf{r})$  be the joint probability of  $\mathbf{x}$  and  $\mathbf{r}$  returned by  $A_{\text{Prob}}$ . Under the intervention  $do(\mathbf{r}_S = \mathbf{x}_{i,S})$ , the features in  $S$  are fixed to  $\mathbf{x}_{i,S}$  and the features in  $\mathcal{F} \setminus S$  remain distributed according to  $p(\mathbf{r}_{\mathcal{F} \setminus S})$ . Therefore, the expected value of  $f(\mathbf{r})$  under this intervention is

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

By the definition of  $v^{(i)}(S)$ ,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

The conclusion follows, linking the  $p$ -SHAP set function to the intervention stated in the theorem.  $\square$

This proof clarifies the interpretation of the attribution contribution:  $v^{(i)}(S)$  measures how the expected value of  $f(\mathbf{r})$  changes when features in  $S$  are set to  $\mathbf{x}_{i,S}$ , i.e., under the intervention  $do(\mathbf{r}_S = \mathbf{x}_{i,S})$ . This intervention is independent of any predefined joint probability distribution  $p(\mathbf{x}, \mathbf{r})$ . Thus  $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$  combines the base expected counterfactual output with the intervention represented by the attribution term.

## E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

**Theorem E.1** (Theorem 5.1 in the main text). *Let  $\mathbf{q} \in \mathbb{R}^{n \times d}$  be the aligned reference matrix derived from the counterfactual set  $\mathbf{r} \in \mathbb{R}^{m \times d}$  via the value assignment step  $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ , where  $\mathbf{p}$  is the coupling matrix. For any dimensions  $n, m \geq 1$ , the refined counterfactual  $\mathbf{z}$  constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

<sup>5</sup>Note that  $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ . Because both  $\pi$  and  $\mathbf{p}_{\text{OT}}$  denote joint probability of  $\mathbf{x}$  and  $\mathbf{r}$ , however,  $\pi$  makes the summation the minimum for the  $W_1$  term across all possible joint distributions, whereas  $\mathbf{p}_{\text{OT}}$  does not.

This inequality guarantees that the refined counterfactual  $\mathbf{z}$  is strictly closer (or equal) to the factual input  $\mathbf{x}$  compared to the aligned counterfactual proposal  $\mathbf{q}$ , ensuring that the refinement process introduces no additional divergence.

*Proof.* Let  $\mathbf{q}$  be the aligned reference matrix generated by  $A_{\text{value}}(\mathbf{r}, \mathbf{p})$ . The proof formalizes why COLA's refinement cannot add distance beyond the aligned proposal. If  $c_{ik} = 1$ , the element is modified and  $z_{ik} = q_{ik}$ ; if  $c_{ik} = 0$ , the element is left factual and  $z_{ik} = x_{ik}$ . Hence

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms for the aligned proposal  $\mathbf{q}$  and the refined counterfactual  $\mathbf{z}$  are

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds because of equation (12).

Because the difference  $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$ , the refined point is no farther from the factual data than the aligned proposal:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots gives the claimed proximity guarantee:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

Taking square roots proves the theorem and makes explicit that COLA's sparse editing step only removes proposal changes; it never adds new changes outside the aligned proposal.  $\square$

**Corollary E.2** (Corollary 5.2 in the main text). *In the special case where  $n = m$  and the OT plan  $\mathbf{p}$  corresponds to a deterministic permutation  $\sigma$  (i.e., obtained without entropic regularization,  $\varepsilon = 0$ ), the matrix  $\mathbf{q}$  becomes a row-permuted version of  $\mathbf{r}$  (denoted as  $\mathbf{r}_\sigma$ , where  $q_i = r_{\sigma(i)}$ ). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

*recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by  $\sigma$ .*

*Proof.* Under  $n = m$  and  $\varepsilon = 0$ , the optimal transport plan  $\mathbf{p}$  is a permutation matrix associated with a bijection  $\sigma$ . The value assignment algorithm  $A_{\text{value}}$ , whether maximizing or averaging, therefore assigns  $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$  for every instance  $i$ . Substituting the permuted matrix  $\mathbf{r}_\sigma$  for  $\mathbf{q}$  in Theorem 5.1, namely  $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$ , directly yields the corollary.  $\square$

## F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

This section supports the scalability claim by separating the cost of each COLA component. The analysis shows how the refinement overhead depends on the CE generator, the coupling computation, Shapley estimation, value assignment, and the final sparse-edit step.

First, consider  $A_{\text{Prob}}$ , the component that makes factual-counterfactual alignment explicit. If the alignment between  $\mathbf{x}$  and  $\mathbf{r}$  is known a priori,  $A_{\text{Prob}}$  constructs  $\mathbf{p}$  from that knowledge in  $\mathcal{O}(n \times m)$ . Otherwise, the OT instantiation computes  $\mathbf{p}$  with the Sinkhorn–Knopp algorithm in  $\mathcal{O}(n \times m \times \log(1/\varepsilon))$ .

Next, consider  $A_{\text{Shap}}$ , which provides the attribution signal used to spend the edit budget. For each sampled subset, the model is evaluated on all  $n$  data points, giving  $\mathcal{O}(n)$  evaluations per subset. Incorporating reference values from the reference table  $\mathbf{r}$  requires replacing selected feature values by baseline values; this access is  $\mathcal{O}(m)$  in the table but, with precomputed reference access, is treated as  $\mathcal{O}(d)$ . With  $M_{\text{Shap}}$  sampled subsets, the resulting complexity is  $\mathcal{O}(n \times d \times M_{\text{Shap}})$ .

The normalization in line 4 takes  $\mathcal{O}(n \times d)$  and turns attribution scores into the sampling policy for sparse edits.

For value assignment,  $A_{\text{Value}}^{\text{max}}$  and  $A_{\text{Value}}^{\text{avg}}$  have different costs. For  $A_{\text{Value}}^{\text{max}}$ , each row  $\mathbf{x}_i$  requires scanning the probabilities  $p_{ij}$  for all  $j \in \{1, 2, \dots, m\}$ , identifying the highest-probability reference row  $\mathbf{r}_j$ , and assigning  $q_{ik} = r_{\tau(i),k}$  where  $\tau(i) = \arg \max_j p_{ij}$ . These steps cost  $\mathcal{O}(m)$ ,  $\mathcal{O}(m)$ , and  $\mathcal{O}(d)$  per row, so the total cost is  $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (m + d))$ .

For  $A_{\text{Value}}^{\text{avg}}$ , each row  $\mathbf{x}_i$  requires computing probabilities  $p_{ij}$ , computing the row sum  $\sum_{j'=1}^m p_{ij'}$ , and forming the weighted average  $q_{ik} = \sum_{j=1}^m \left( \frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$ . The first two steps cost  $\mathcal{O}(m)$  each per row, and the weighted average costs  $\mathcal{O}(m \times d)$  per row. Across  $n$  rows, the total is  $\mathcal{O}(n \times m \times d)$ .

For lines 6–16, selecting and applying the sparse action mask costs  $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$  because  $C \leq n \times d$ .

Combining the components, COLA with  $A_{\text{Value}}^{\text{max}}$  has complexity

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and COLA with  $A_{\text{Value}}^{\text{avg}}$  has complexity

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Thus, with respect to  $n$ ,  $m$ ,  $d$ , and entropic-OT regularization parameter  $\varepsilon$ , the refinement overhead is summarized as

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where  $N = \mathcal{O}(nm) + \mathcal{O}(nd)$  for  $A_{\text{Value}}^{\text{max}}$  and  $N = \mathcal{O}(nmd)$  for  $A_{\text{Value}}^{\text{avg}}$ .

## G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

This section provides the exact benchmark used to evaluate the near-optimality contribution. We assume a known factual-counterfactual alignment and solve equation (1) with  $D$  set to MeanD; this benchmark is the exact counterpart of equation (1) used in the main results:

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with  $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$ . Since COLA is used,  $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$  and the refined  $\mathbf{z}$  stays closer to  $\mathbf{x}$  than  $\mathbf{r}$ ; therefore equation (1b) is omitted in this benchmark. The resulting problem is

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

The original constraints equation (1d) and equation (1e) collapse into equation (13c) because CF- $p_{\text{Ect}}$  supplies an exact reference for each factual row. For each  $\mathbf{x}_i$ , the aligned  $\mathbf{r}_j$  is used in both  $A_{\text{Shap}}$  and  $A_{\text{Value}}$ , so each feature  $x_{ik}$  either remains unchanged or is replaced by its aligned counterfactual value  $r_{jk}$ . Thus  $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$ , with the binary variable  $c_{ik}$  determining the edit.

Because the alignment between each  $\mathbf{x}_i$  and its corresponding  $\mathbf{r}_j$  is known,  $\mathbf{q}$  is fixed; both  $A_{\text{Values}}^{\max}$  and  $A_{\text{Values}}^{\text{avg}}$  return the same  $\mathbf{q}$ . For any data point  $i$  and feature set  $S \subseteq \mathcal{F}$ , let  $\mathbf{z}_{iS}$  denote the candidate in which all features  $k \in S$  are changed to  $q_{ik}$  and all features  $h \in \mathcal{F} \setminus S$  remain  $x_{ih}$ . The set of all  $\mathbf{z}_{iS}$  over  $S \subseteq \mathcal{F}$  enumerates the possible values of  $\mathbf{z}_i$ . Define the corresponding scalar

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

For any  $\mathbf{z}_i$  in equation (13), a binary variable  $a_{iS}$  selects one feature subset  $S$ ; this selection form is equivalent to equation (13). The term  $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$  can be represented as

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The exact benchmark is therefore the following mixed-integer program. It keeps the distinction between the factual  $\mathbf{x}$ , the generated counterfactual  $\mathbf{r}$ , and the refined candidate  $\mathbf{z}$  explicit:  $\mathbf{r}$  supplies allowable replacement values, while  $\mathbf{z}$  records the selected sparse realization of those values. The program then optimizes over  $\mathbf{z}$  given this exact alignment with  $\mathbf{r}$ .

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing  $\eta$  under equation (14b) and equation (14c) is equivalent to minimizing the absolute objective in equation (13). Constraint equation (14d) ensures that each data point  $i$  receives exactly one feature-modification plan  $a_{iS}$ , and equation (14e) enforces the action budget from equation (13b). Solving equation (14) gives the optimality benchmark for COLA under known alignment, reported in Figure 4 of Section 6.

## H EXTENDED NUMERICAL RESULTS

The extended results reinforce the main empirical contribution: counterfactual-aware and alignment-aware attribution is necessary for sparse CE refinement. In Figures 5–8 and the additional panels in Figures 5–8, CF- $p_{\text{Uni}}$  generally improves over RB- $p_{\text{Uni}}$ , and CF- $p_{\text{OT}}$  consistently improves over RB- $p_{\text{OT}}$  when the methods differ only in  $A_{\text{Shap}}$ . Thus *RB-SHAP is not suitable for FA in CE* when the goal is sparse counterfactual action.

Lines 3–4 are therefore the decisive ablation point in COLA. The paired comparisons show that replacing RB-SHAP with a counterfactual-aware Shapley rule improves the quality of the selected edits, and the improvement is strongest when the Shapley rule uses the OT coupling.

Line 5 supplies the other half of the contribution: the joint probability  $p(\mathbf{x}, \mathbf{r})$  constructs  $\mathbf{q}$ , which tells the factual  $x$  which direction to change its features in order to move toward the target model outcome. Figures 5–8 show that CF- $p_{OT}$  consistently outperforms the other methods. CF- $p_{Uni}$ , CF- $p_{Rnd}$ , and CF- $p_{OT}$  all propose joint probabilities  $\mathbf{p}$  when exact factual-counterfactual alignment is unknown, but their results differ substantially. This supports the claim that knowing the CE marginal distribution is insufficient; the alignment itself is crucial.

The extended experiments therefore sharpen the role of OT: it is especially useful when factual-counterfactual alignment is missing or inaccurate. Even when a CE algorithm explicitly matches each factual instance to a counterfactual, that pairing is not necessarily optimal for attribution or sparse refinement, as supported by Figure 4 in Section 6.

Importantly,  $p_{OT}$  need not be the true data-generating joint distribution of  $\mathbf{x}$  and  $\mathbf{r}$ . Its role is operational: it guides COLA in using  $\mathbf{x}$  and  $\mathbf{r}$  jointly for both FA and CE. The QDA column in Figure 5 further shows that OT-based methods remain stable while other methods can diverge from the target. COLA is nevertheless *not limited to using OT* as  $A_{Prob}$ . Since Figure 4 still shows a non-negligible gap to global optimality for known alignments, designing better  $A_{Prob}$  algorithms and better choices of  $\mathbf{p}$  remains a natural future direction.

## I EXPERIMENTS REPRODUCIBILITY

The experiments were run on a high performance computing (HPC) cluster, with four nodes used in parallel for the four datasets. Each node has two Intel Xeon Processor 2660v3 CPUs (10 core, 2.60GHz) and 128 GB memory. Runtime is approximately 5–10 hours per node depending on dataset size. The experiments can also be reproduced on a laptop, but generally require more time than on the HPC cluster.

For all four datasets, numerical features are standardized and categorical features use either label encoding or one-hot encoding. We did not observe a substantial difference between these two categorical encodings for COLA’s performance. The train-test split is 7 : 3.

The optimality baseline in Figure 4 is solved with Gurobi 11.0.2 ([gur](#)); reproducing that baseline requires a Gurobi license. Open-source operations-research libraries such as Google-OR tools ([goo](#)) can be used as alternatives. The MILP in Appendix G is computationally expensive, so it is practical only for small-scale settings such as German Credit. To compute the optimality baseline on other datasets, the feature set must be reduced.

Model configurations are fixed across the evaluation to keep the empirical contribution focused on refinement rather than model tuning. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA use scikit-learn implementations ([skl](#)) with default hyperparameters. DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN has three layers and the SVM uses a linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) use their scikit-learn interfaces with default hyperparameters.

## THE USE OF LARGE LANGUAGE MODELS (LLMs)

Large language models (LLMs) were used only as general-purpose writing-assistance tools. Their role was limited to grammar, wording, and light copy-editing of author-written text. They did not contribute research ideas, modeling choices, experimental design, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors. Any LLM-suggested phrasing was reviewed and edited before inclusion. The authors take full responsibility for all content; LLMs are not authors or contributors. This disclosure follows the ICLR policy on LLM usage.

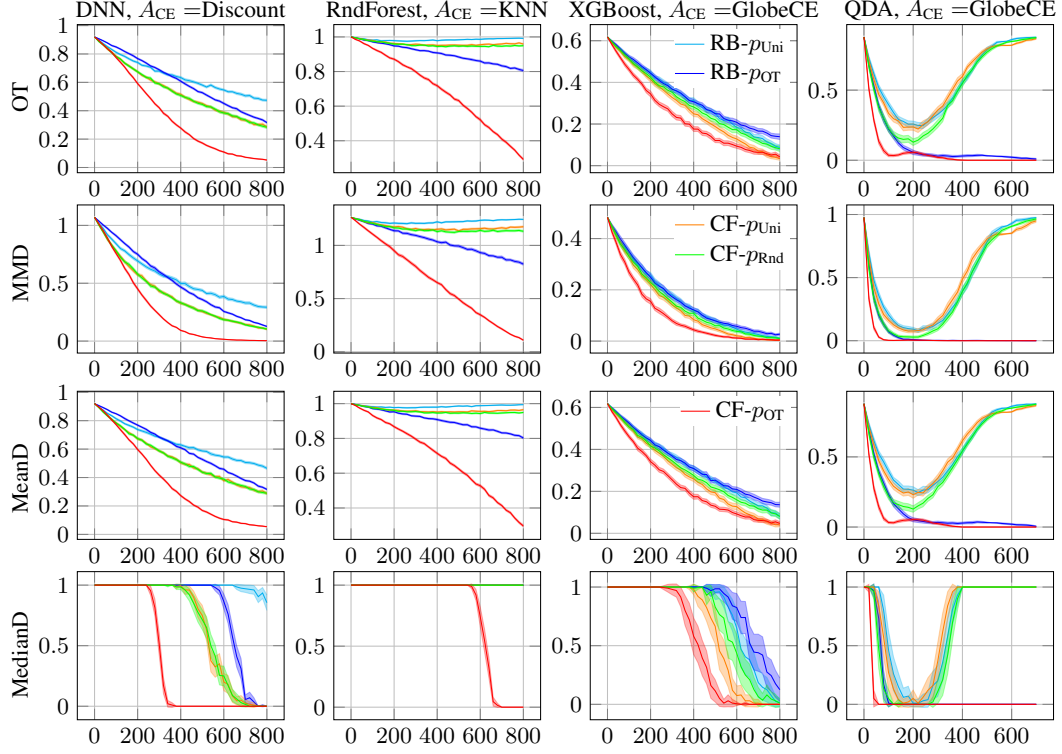


Figure 5: [HELOC] Extended sparse-refinement curves for HELOC:  $D(f(\mathbf{z}), \mathbf{y}^*)$  versus allowed actions  $C$ . Each experiment uses 100 runs; shaded regions show 99.9% confidence intervals; legends apply to all plots.  $A_{\text{Value}}^{\text{avg}}$  is used.

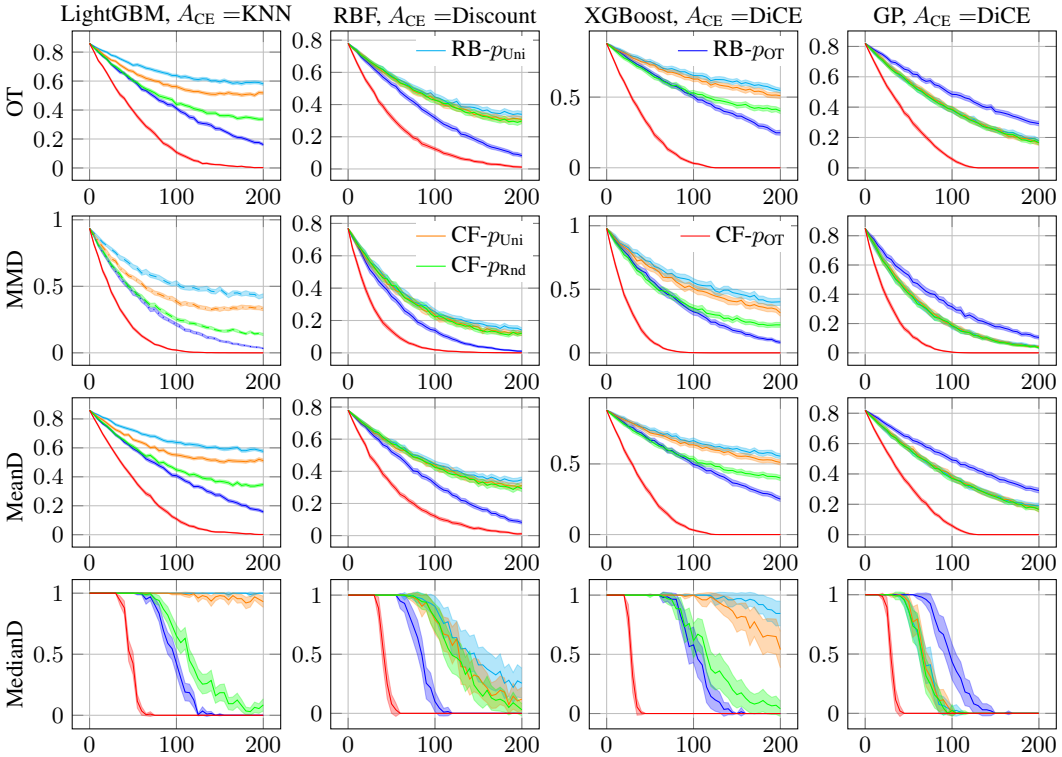


Figure 6: [German Credit] Extended sparse-refinement curves for German Credit:  $D(f(\mathbf{z}), \mathbf{y}^*)$  versus allowed actions  $C$ . Each experiment uses 100 runs; shaded regions show 99.9% confidence intervals; legends apply to all plots.  $A_{\text{Value}}^{\text{avg}}$  is used.

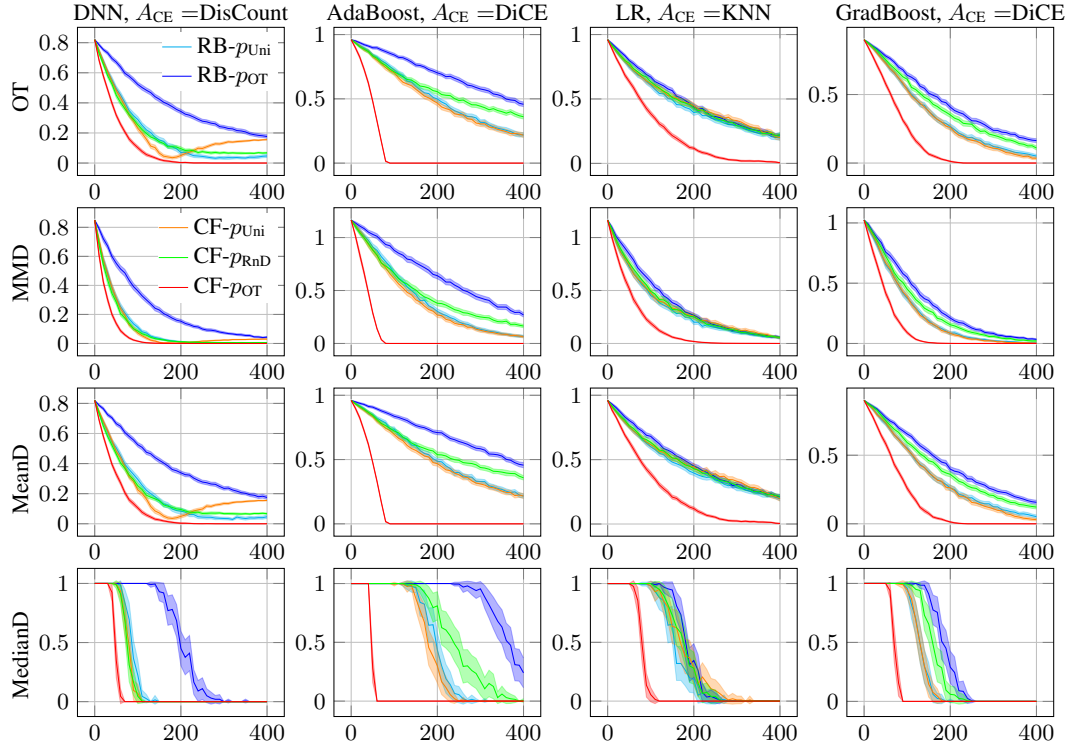


Figure 7: [Hotel Bookings] Extended sparse-refinement curves for Hotel Bookings:  $D(f(\mathbf{z}), \mathbf{y}^*)$  versus allowed actions  $C$ . Each experiment uses 100 runs; shaded regions show 99.9% confidence intervals; legends apply to all plots.  $A_{\text{Value}}^{\max}$  is used.

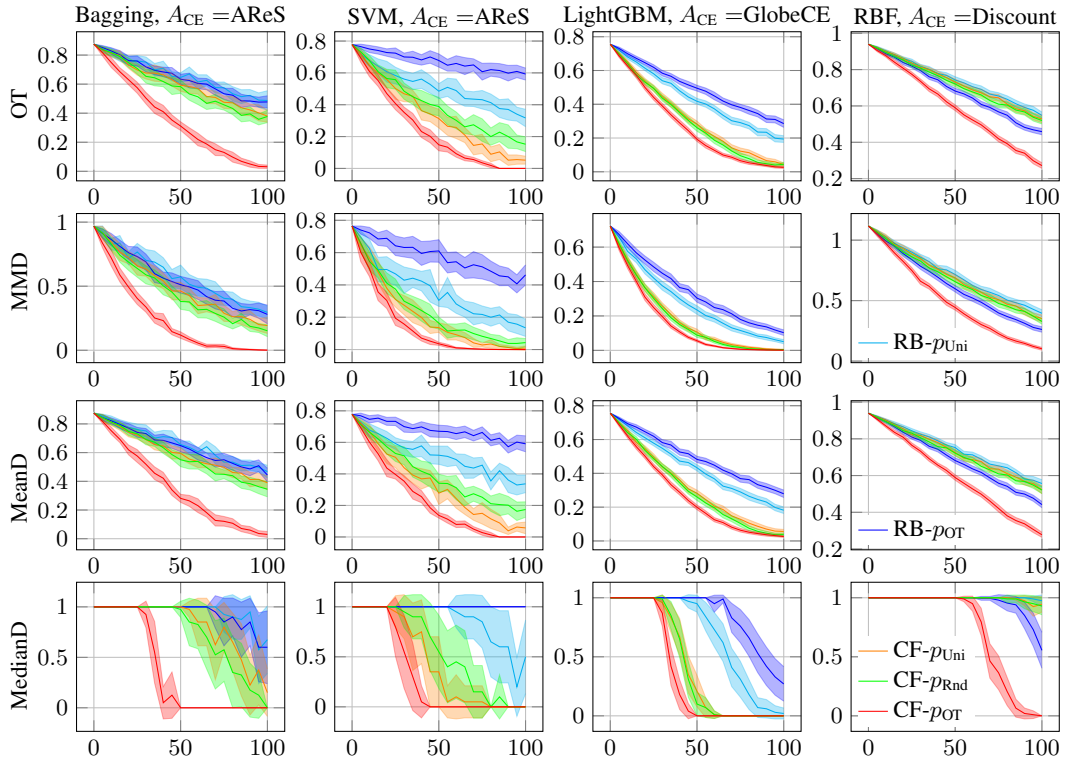


Figure 8: [COMPAS] Extended sparse-refinement curves for COMPAS:  $D(f(\mathbf{z}), \mathbf{y}^*)$  versus allowed actions  $C$ . Each experiment uses 100 runs; shaded regions show 99.9% confidence intervals; legends apply to all plots.  $A_{\text{Value}}^{\max}$  is used.